

Kaylee Y. Vo

Curriculum Vitae

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Education

- 2024–2026 **Master of Liberal Arts in Extension Studies, Data Science**, *Harvard University*, Cambridge, MA, GPA: 4.0/4.0
Fields: Scientific Machine Learning, Applied Mathematics
Capstone: "Trajectory-Aware Prompt Optimization for AI Agents"
- 2012–2016 **B.A., Economics**, *Stanford University*, Stanford, CA, GPA: 3.4/4.0
Minor: Statistics

Research Experience

- August 2026 - **Research Data Scientist**, *Sandia National Laboratories*, Albuquerque, NM
Present Advisor: Dr. Mary Beth Mulcahy, PhD
(Incoming) Project: Scientific ML and AI Interpretability
- 2026 - **Research Intern**, *EleutherAI*, Seattle, WA
Present Advisor: Goncalo Paulo PhD
Project: Benchmarks for Causal Steering using Natural Language Autoencoders
- 2025–2026 **Research Associate**, *Harvard John A. Paulson School of Engineering and Applied Sciences*, Cambridge, MA
Advisor: Prof. Pavlos Protopapas
Project: Variational Boosting for Physics-Informed Neural Networks
- 2016–2017 **Research Associate**, *Federal Reserve Bank of San Francisco*, San Francisco, CA
Advisor: Dr. Adam Shapiro
Project: Using Brexit to Identify the Nature of Price Rigidities
- 2015–2016 **Research Fellow**, *Stanford Center on Poverty and Inequality*, Stanford, CA
Advisors: Prof. Mark Duggan, Prof. David Rogosa, Dr. Charles Varner
Project: The Effect of Prescription Drug Monitoring Programs on the Opioid Epidemic

Awards, Fellowships, and Patents

- 2026 **Harvard University Dean's List Academic Achievement Award**
- 2019 **Patent #11562425**: System for automated origination of capital based on point-of-sale data informed by location
- 2019 **Patent #11100575**: System for automated origination of capital based on point-of-sale data informed by time of year

- 2019 **Patent #10956974:** Dynamic origination of capital pricing determination based on forecasted point-of-sale revenue
- 2012–2016 **Stanford University Academic Scholarship** (full-ride)
- Fall 2015 **Stanford Economics Student of the Month**
- Summer 2015 **Stanford Undergraduate Grant for Economics Research**
- Summer 2015 **Stanford Haas Center for Public Service Research Grant**
- 2015–2016 **Undergraduate Research Fellow Award**, Center on Poverty and Inequality

Professional Experience (Non-Research)

- Mar **Senior Software Engineer, Data Engineering, Nuna Inc., San Francisco, CA**
- 2023–May 2024
 - Productionized the Chronic Illness and Disability Payment System (CDPS) model. The model was deployed to EMR Spark clusters, using Prefect for task orchestration. Optimized the CDPS algorithm, resulting in an 80% reduction in runtime.
 - Built data transformations to map external client data to Nuna's data schema. Schemas are defined using Avro files and written to an AWS Glue. Performed Spark optimizations to reduce the ETL runtime. Wrote unit and integration tests to ensure data quality. The work culminated in a production Glue database used across the company.
- Nov **Senior Data Scientist, Carbon Health, San Francisco, CA**
- 2022–Mar 2023
 - Built DBT transformations on top of Snowflake to procure data for Looker dashboards. Dashboards were used to measure provider efficiency, with metrics such as risk-adjusted total spending per patient per month.
- Nov **Data Scientist, Pngme**
- 2021–Nov 2022
 - Developed an MVP loan default prediction model. Model was a logistic regression fitted on cross-sectional data. Features are financial metrics transformed into weight-of-evidence features using the OptBinning library. Lift, profit curves, and Shapley values were used as monitoring metrics, and AUC was the performance metric.
 - Lead feature engineering and modeling R&D work to ensure ML model performance increases commensurately with data growth. We generated custom word embeddings with BERT using user SMS data as input. The final model was a mixture of experts, with a logistic regression meta-model. The final model was customer-facing and deployed to production.
- Nov **Data Scientist, Toast Inc., Boston, MA**
- 2018–Apr 2020
 - Built a new line of business (LOB) called Toast Capital. The product comprises three patented models: LSTM Revenue Prediction (patent #11562425), a probability of default model (patent #11100575), a dynamic pricing engine (patent #10956974), an ETL data pipeline, and API endpoints. Total origination was over \$15 million, the second most profitable line of business.
 - Designed prototypes to improve the PD model. This includes model selection, model stability, model parsimony, TTC PD vs PIT PD, and testing regression assumptions. I also led an R&D initiative to develop a generalized PD model.

- Jun **Data Scientist**, *Moody's Analytics*, New York, NY
- 2017–Oct 2018 ○ Computer Vision: Developed an in-house OCR software using Tesseract and OpenCV to parse text and objects from images. OCR extracted text, tables and numbers from PDF's, augmenting Moody's training data, increasing the PD model's accuracy.
- Sentiment Analysis: Constructed models to determine if sentiment scores improve the predictive power of Moody's default probability model. Utilized a variety of specifications, namely Kalman Filters, VARs, and MIDAS regressions.
- Jul **Research Associate**, *Federal Reserve Bank of San Francisco*, San Francisco, CA
- 2016–May 2017 ○ Implemented an algorithm to estimate the sentiments of the FOMC and the markets, following macroeconomic shocks. I used text transcripts from FOMC meetings and a database of news articles as the test set. The algorithm was trained on a tagged corpus given by SentiWord. Presented project at City College of San Francisco and was featured on Bloomberg News.
- Leveraged R's object-oriented paradigms (S3) to build construct models that use sentiment indexes to improve economic forecasts. I utilized lasso and group lasso shrinkage methods, using AIC and BIC as the model selection criteria.

Teaching Experience

- Jan 2026 Mathematical Modeling (AM115) - Harvard College

Publications

- Sept 2026 Journal of Scientific Computing and Machine Learning, 2026 Conference Proceedings — "Variational Boosting for Physics-Informed Neural Networks"
- Abstract *We introduce a variational boosting framework where solutions are constructed additively in function space. Each stage minimizes the nonlinear residual over an affine subspace of a Hilbert space. Because each correction network is small, the minimization admits conjugate gradient updates.*

Presentations

- Aug 2026 "Trajectory-Aware Prompt Optimization for AI Agents" (*poster*). Agentic AI Summit, UC Berkeley.
- Oct 2026 "Variational Boosting for Physics-Informed Neural Networks" (*oral*). Women in Statistics and Data Science, American Statistical Association.
- Jul–Aug 2026 "Variational Boosting for Physics-Informed Neural Networks" (*poster*). STAI-X, American Statistical Association.
- Nov 2016 "Measuring News Sentiment from FOMC Transcripts" (*invited talk*). San Francisco City College.
- Jun 2016 "The Effect of Prescription Drug Monitoring Programs on the Opioid Epidemic" (*poster*). Stanford Undergraduate Research Symposium.

Skills

Programming	Python (scikit-learn, PyTorch, TensorFlow, JAX), R, SQL (Postgres, Snowflake, Big-Query), Bash, MATLAB
ML & AI	LLMs (LLaMA, BERT), Diffusion Models (U-Net, Stable Diffusion), Computer Vision (R-CNN, ViT, segmentation), Multimodal (CLIP)
Mathematics	Stochastic & Partial Differential Equations, Real & Functional Analysis, Matrix Theory, Causal Inference, Time Series
Infrastructure	AWS (SageMaker, S3, EC2, EMR, Glue), Kubernetes, Spark, Docker, Prefect, Git/GitHub Actions

References

Zhiming Kuang

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School of Engineering and Applied Sciences, Harvard University
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